

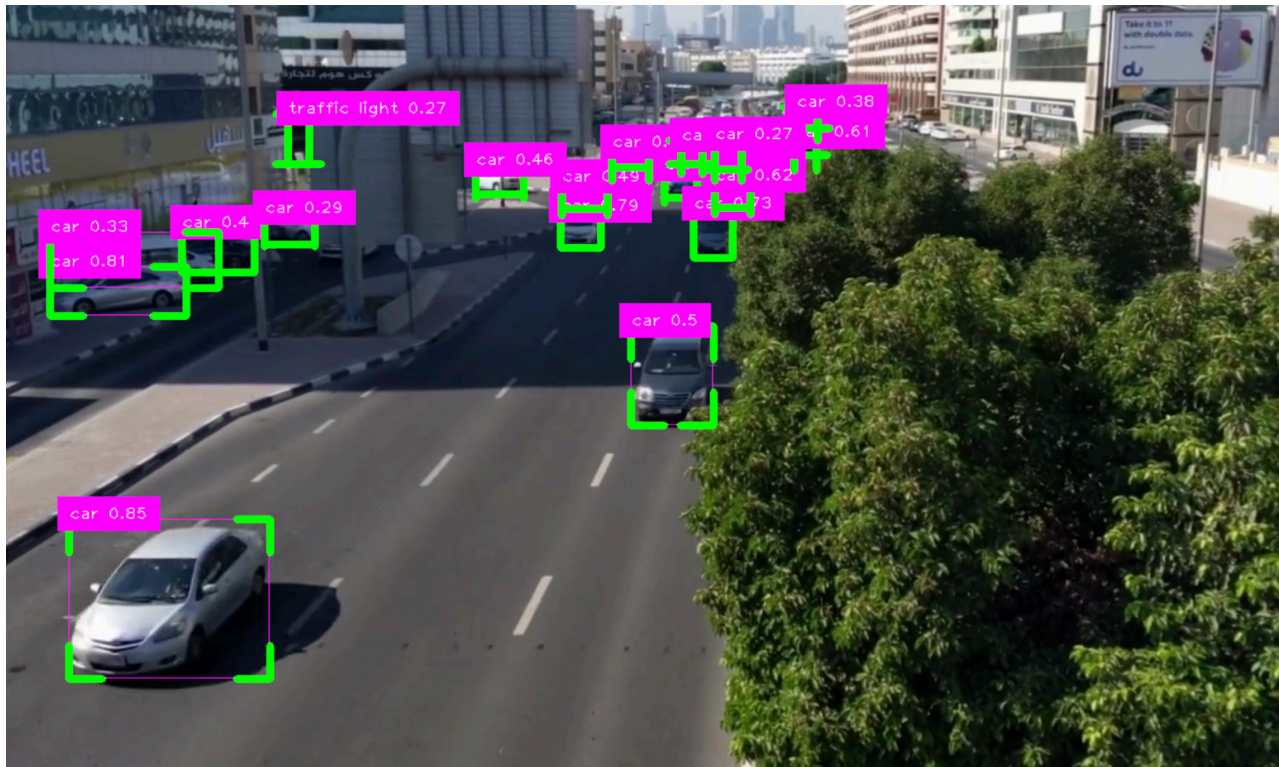
```
#cap = cv2.VideoCapture(cv2.CAP_DSHOW + 0)
#cap.set(cv2.CAP_PROP_FRAME_WIDTH, 1280)
#cap.set(cv2.CAP_PROP_FRAME_HEIGHT, 720)

#cap = cv2.VideoCapture("../Videos/motorbikes.mp4") # For Video
cap = cv2.VideoCapture("../Videos/people.mp4") # For Video
```

YOLO-Car Counter

YOLO-Car-Counter

```
cap = cv2.VideoCapture("../Videos/cars.mp4") # For Video
```



YOLO-Car-Counter

classname0l car, truck, bus, motobike 만 표시한다.

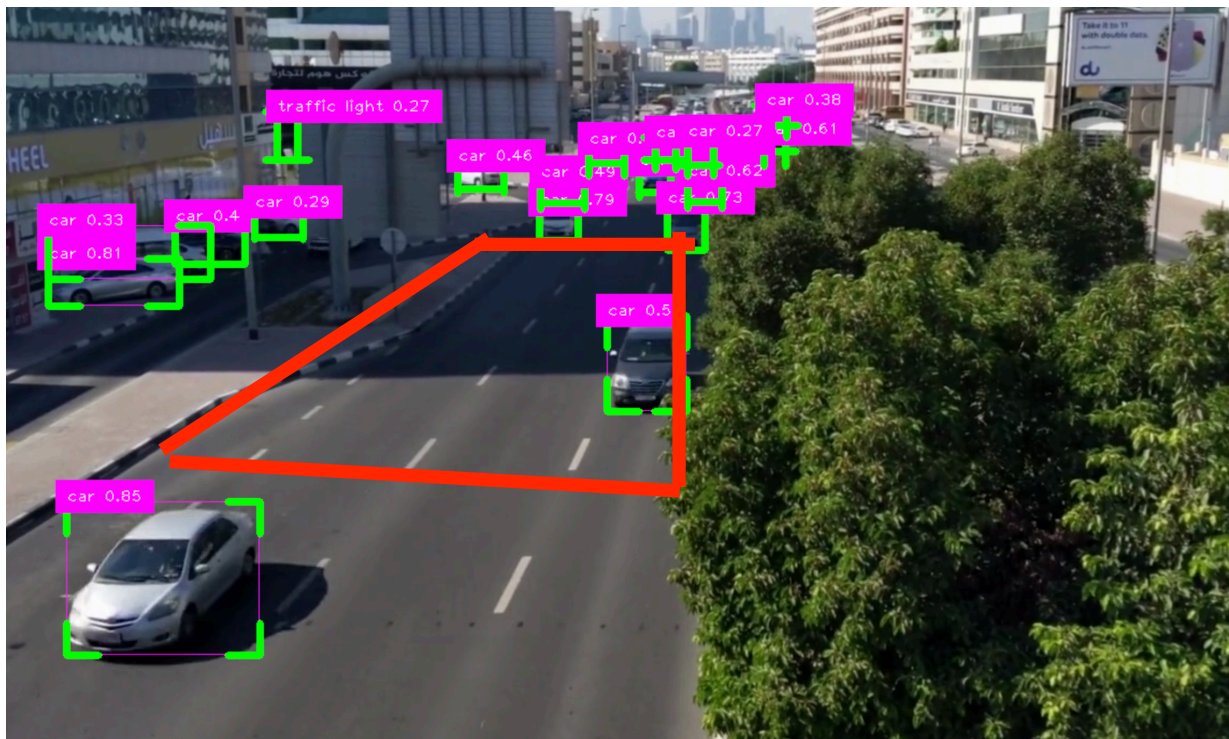
```
cls = int(box.cls[0])

currentClass = classNames[cls]

if (
    currentClass == "car"
    or currentClass == "truck"
    or currentClass == "bus"
    or currentClass == "motobike"
    and conf > 0.3
):
    cvzone.putTextRect(
        img,
        f"{classNames[cls]} {conf}",
        (max(0, x1), max(35, y1)),
        scale=0.8,
        thickness=1,
        offset=3,
    )
    cvzone.cornerRect(img, (x1, y1, w, h), l=9, rt=5)
```

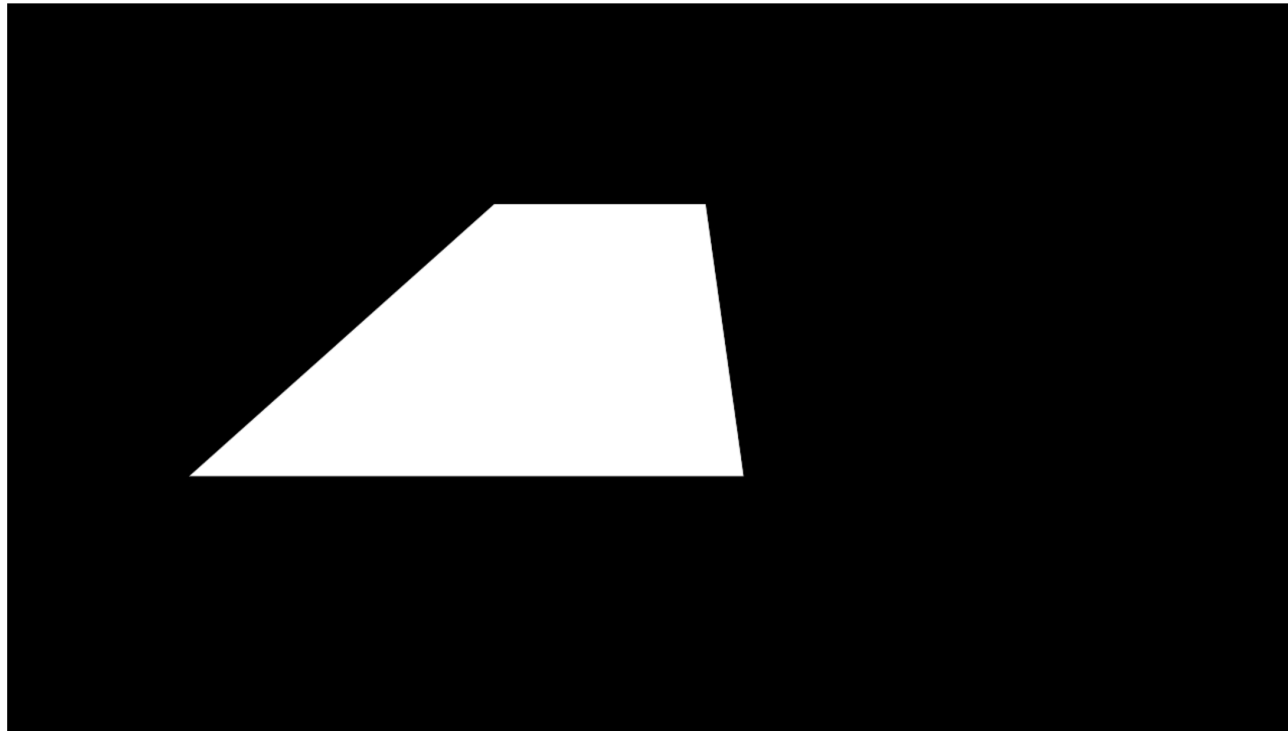

YOLO-Car-Counter

라인을 지나가는 car 를 count 한다.



YOLO-Car-Counter

mask.png



YOLO-Car-Counter



```
mask = cv2.imread("mask.png")
```

```
while True:
```

```
    success, img = cap.read()
```

```
    imgRegion = cv2.bitwise_and(img, mask)
```



```
    results = model(imgRegion, stream=True)
```

```
cv2.imshow("Image", img)
```

```
cv2.imshow("ImageRegion", imgRegion)
```

```
cv2.waitKey(1)
```



```
from sort import *
```

```
while True:
    success, img = cap.read()
    imgRegion = cv2.bitwise_and(img, mask)
    results = model(imgRegion, stream=True)

    detections = np.empty((0, 5))
```



주석



```
    if currentClass == "car" or currentClass == "truck" or currentClass == "bus" \
        or currentClass == "motorbike" and conf > 0.3:
        # cvzone.putTextRect(img, f'{currentClass} {conf}', (max(0, x1), max(35, y1)),
        #                   scale=0.6, thickness=1, offset=3)
        # cvzone.cornerRect(img, (x1, y1, w, h), l=9, rt=5)

        currentArray = np.array([x1, y1, x2, y2, conf])
        detections = np.vstack((detections, currentArray))
```

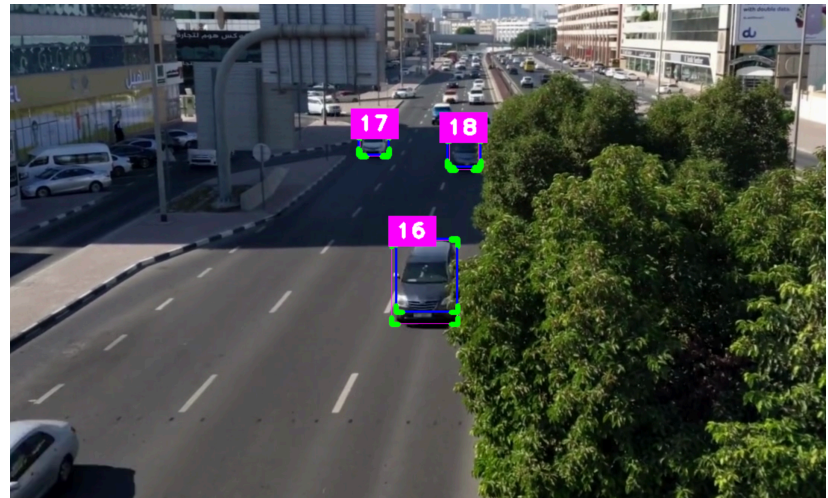




```
resultsTracker = tracker.update(detections)
```



```
for result in resultsTracker:
    x1, y1, x2, y2, id = result
    x1, y1, x2, y2 = int(x1), int(y1), int(x2), int(y2)
    print(result)
    w, h = x2 - x1, y2 - y1
    cvzone.cornerRect(img, (x1, y1, w, h), l=9, rt=2, colorR=(255, 0, 0))
    cvzone.putTextRect(
        img,
        f"{int(id)}",
        (max(0, x1), max(35, y1)),
        scale=2,
        thickness=3,
        offset=10,
    )
```



YOLO-Car-Counter

```
mask = cv2.imread("mask.png")  
limits = [400, 297, 673, 297]
```



```
while True:  
    success, img = cap.read()  
    imgRegion = cv2.bitwise_and(img, mask)
```

```
    imgGraphics = cv2.imread("graphics.png", cv2.IMREAD_UNCHANGED)  
    img = cvzone.overlayPNG(img, imgGraphics, (0, 0))
```



```
cv2.line(img, (limits[0], limits[1]), (limits[2], limits[3]), (0, 0, 255), 5)
```

```
for result in resultsTracker:  
    x1, y1, x2, y2, id = result
```

